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Lab Exercise 01 Report



FitFlow Redesign

User Research – Lab Exercise 01 Report

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Case Study	FitFlow Redesign Scenario

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Case Study Overview – FitFlow Redesign

FitFlow is a fitness-tracking app from a mid-sized health-tech startup that had been steadily losing users over the past year. App-store ratings fell from an average of 4.6 to 3.8 stars and user retention dropped sharply. Analysis of early usage data showed that 68% of new users abandoned the app shortly after onboarding, and many found nutrition logging tedious. In response, the product team – led by Product Manager Priya Sharma – launched a structured, human-centred user-research initiative to uncover pain points and validate three proposed features: AI-powered personalised workouts, social community features, and improved nutrition tracking. This report documents that research across Activities 1–6.

Key Figures and Percentage Details

Indicator	Before Redesign	After Redesign
Average app-store rating	3.8 stars (down from 4.6)	Recovered to ~4.5 stars
Users abandoning after onboarding	68%	Significantly reduced
User retention (past year)	Declining	Improved by ~35%
Usability score (SUS)	68 / 100	87 / 100
Beta testers involved	—	150 users

These figures explain why the redesign was needed: the average rating fell by 0.8 stars (about a 17% decline) and 68% of new users left shortly after onboarding, signalling serious usability and retention problems. The user research documented in this report was carried out to understand the causes behind these numbers and to guide the redesign that later improved retention by roughly 35% and raised the usability (SUS) score from 68 to 87.

Activity 1: Stakeholder Identification

Stakeholders for the FitFlow redesign are grouped into primary (direct users of the app), secondary (people who influence or support those users), and tertiary (organisational and indirectly-affected parties). The table below lists each stakeholder, a short description, and why they matter to the research.

Category	Stakeholder	Description	Importance for Research
Primary	Current FitFlow users	Existing users who actively use the fitness-tracking features.	Give feedback on current usability issues, satisfaction, and desired improvements.
	Lapsed users	Users who have stopped using FitFlow.	Help identify reasons for drop-off and retention problems.
	Fitness beginners	New users who need guidance and motivation.	Reveal onboarding difficulties and the need for personalised support.
	Fitness enthusiasts	Experienced users tracking workouts and progress.	Provide insight into advanced features and performance-tracking needs.
	Users with different goals	Users focused on weight loss, strength, health, or wellness.	Inform flexible, personalised workout and nutrition features.
Secondary	Family members	People who may interact with social / community features.	Clarify social-support and accountability requirements.
	Fitness trainers	Professionals who guide users' workouts.	Offer expert views on workout recommendations and user behaviour.
	Healthcare providers	Professionals interested in anonymised wellness trends.	Advise on health-data usage and privacy requirements.
	Gym partners	External fitness organisations collaborating with FitFlow.	Identify partnership opportunities and engagement strategies.
	Online fitness communities	Groups that influence fitness habits and motivation.	Reveal social-engagement expectations.
Tertiary	Product Manager (Priya Sharma)	Leads product direction and redesign decisions.	Ensures research outcomes align with business goals.
	UX / UI designers	Design the interfaces and user experience.	Use findings to create user-friendly designs.

Category	Stakeholder	Description	Importance for Research
	Software engineers	Build and maintain the application.	Need technical requirements and implementation insight.
	Data science & AI team	Build the AI workout and personalisation features.	Need data requirements and AI expectations.
	CEO & business leaders	Focus on revenue recovery and product success.	Ensure the redesign supports business objectives.
	Marketing team	Promote the redesigned application.	Use insights to craft effective campaigns.
	Regulatory bodies	Ensure privacy and compliance requirements.	Ensure ethical handling of user information.

Activity 2: Select Suitable Research Methods

Selected Research Methods

1. Interviews – to collect detailed opinions, emotions, and experiences from users.
2. Survey – to gather quantitative data from many users about satisfaction, usage patterns, and problems.
3. Observations – to see how users naturally interact with the app during workouts and daily tracking.
4. Contextual inquiry – to observe users in their real environment while completing fitness activities.

Justification

A mixed-method approach suits FitFlow because it combines quantitative and qualitative data. Surveys and app analytics identify common problems across a large user group, such as onboarding difficulties and nutrition-logging issues. Interviews provide deeper insight into user emotions, motivations, and frustrations. Observation and contextual inquiry reveal real-world behaviour during workouts and tracking. Together these methods build a complete understanding of user needs and support evidence-based redesign decisions.

Combining Qualitative and Quantitative Approaches

Approach	Methods	Expected Outcome
Quantitative research	Surveys, Firebase analytics, Mixpanel analysis	Identify patterns, user-satisfaction levels, and retention issues.
Qualitative research	Interviews, observations, contextual inquiry	Understand user emotions, motivations, and pain points.

Activity 3: Design Research Instruments

Semi-Structured Interview Guide

Participant Information

- Age:
- Fitness experience:
- Frequency of using fitness apps:

Interview Questions

5. How do you currently use fitness-tracking applications?
6. What features of FitFlow do you find most useful?
7. What problems or frustrations have you experienced while using FitFlow?
8. Describe your experience during the onboarding process.
9. How do you feel about the current workout recommendations provided by FitFlow?
10. Would an AI-powered personalised workout plan improve your experience? Why?
11. What type of social features would motivate you to stay active?
12. What difficulties do you face when tracking your nutrition?
13. How would you prefer FitFlow to provide progress feedback?
14. What new features would encourage you to continue using FitFlow?

Survey Questionnaire

Section 1: User Background

1. What is your age group?

- ☐ Below 18
- ☐ 18–25
- ☐ 26–35
- ☐ Above 35

2. How often do you use fitness applications?

- ☐ Daily
- ☐ Several times a week
- ☐ Occasionally
- ☐ Rarely

3. How long have you used FitFlow?

- ☐ Less than 1 month
- ☐ 1–6 months
- ☐ More than 6 months

Section 2: App Experience (1 = Very dissatisfied, 5 = Very satisfied)

4. How satisfied are you with FitFlow currently?

- ☐ 1 2 3 4 5

5. How easy is it to complete daily tracking tasks?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

6. What problems have you experienced while using FitFlow? (select all that apply)

- ☐ Difficult onboarding
- ☐ Generic workout plans
- ☐ Difficult nutrition logging
- ☐ Lack of motivation
- ☐ Poor social interaction

7. How useful would an AI personalised workout plan be?

- ☐ Very useful
- ☐ Useful
- ☐ Neutral
- ☐ Not useful

8. Which AI features would you like? (select all that apply)

- ☐ Adaptive workout plans
- ☐ Exercise recommendations
- ☐ AI fitness coach
- ☐ Progress predictions

9. Would you use social community features?

- ☐ Yes
- ☐ No
- ☐ Maybe

10. Which social features interest you most? (select all that apply)

- ☐ Fitness challenges
- ☐ Friends / groups
- ☐ Leaderboards
- ☐ Sharing achievements

11. How often do you track your nutrition?

- ☐ Every day
- ☐ Sometimes
- ☐ Rarely
- ☐ Never

12. Would camera-based food recognition improve nutrition tracking?

- ☐ Yes
- ☐ No
- ☐ Maybe

13. What is the biggest improvement FitFlow needs? (open-ended)

☐

14. What motivates you to continue using fitness apps? (open-ended)

☐

15. Any additional suggestions for the FitFlow redesign? (open-ended)

☐

Observation / Contextual Inquiry Checklist

Observation Area	Checklist
User onboarding	☐ User understands the registration process.
Navigation	☐ User can easily find workout features.
Workout tracking	☐ User completes workout logging successfully.
Nutrition tracking	☐ User finds food logging easy / difficult.
Error handling	☐ User understands system messages.
AI suggestions	☐ User trusts AI recommendations.
Social features	☐ User understands community interactions.
Emotional response	☐ User shows frustration / confusion / satisfaction.
Time taken	☐ Record completion time for tasks.
Suggestions	☐ Collect improvement ideas.

Activity 4: Pilot Test the Instruments with Classmates

Participant	Background	Instrument Tested
Participant 01	University student who uses fitness applications	Survey questionnaire
Participant 02	Regular gym user	Interview guide
Participant 03	Beginner fitness-app user	Survey + observation checklist

Feedback Received During Pilot Testing

Area Evaluated	Feedback
Question clarity	Some questions were too technical and needed simpler wording.
Length	The survey contained slightly repetitive questions and could be shortened.
Difficulty	Participants understood most questions but needed examples for the AI features.
Relevance	Participants suggested adding more questions about motivation and social features.
Time required	The interview took around 15 minutes and the survey 5–7 minutes.

Changes Made After Pilot Testing

Original Issue	Improvement Made
AI-related questions were unclear.	Added simple explanations about AI workout recommendations.
Some survey questions were repetitive.	Removed unnecessary repeated questions.
Limited questions about motivation.	Added questions about user engagement and fitness goals.
Observation checklist lacked emotional tracking.	Added user frustration, confusion, and satisfaction observations.
Interview questions were too broad.	Made questions more specific to FitFlow features.

Pilot Testing Summary

The pilot test improved the quality and usability of the research instruments. Feedback from classmates was used to simplify questions, reduce ambiguity, and ensure the instruments captured important information about FitFlow's usability problems, user expectations, and redesign opportunities. After the modifications, the instruments were considered suitable for conducting user research.

Activity 5: Conduct User Research with at Least 5 Participants

Participant Recruitment

Five simulated participants were selected from two stakeholder groups:

- Primary stakeholders – current fitness-app users and beginner fitness users.
- Secondary stakeholders – fitness trainers and gym instructors.

Research Participants

ID	Stakeholder Group	Profile	Research Method
P01	Primary user	25-year-old university student using fitness apps weekly.	Interview + Survey
P02	Primary user	32-year-old working professional with an irregular workout schedule.	Interview + Survey
P03	Primary user	27-year-old beginner interested in weight management.	Survey + Observation
P04	Secondary user	Fitness trainer who helps clients create workout plans.	Interview
P05	Secondary user	Gym instructor experienced with fitness-tracking tools.	Interview + Observation

Raw Research Notes

Participant P01 – University Student

Key comments

- Finds the current workout plans repetitive.
- Wants more motivation features.
- Interested in fitness challenges with friends.

Pain points

- Lack of personalisation.
- Difficulty maintaining consistency.

Feature suggestions

- AI workout recommendations.
- Achievement badges.
- Social competitions.

Participant P02 – Working Professional

Key comments

- Has limited time due to a busy work schedule.
- Cannot follow fixed workout plans.

Pain points

- Workout plans do not adapt to available time.

- Tracking progress requires too much effort.

Feature suggestions

- AI-generated flexible workouts.
- Quick 15-minute workout options.
- Smart reminders.

Participant P03 – Beginner Fitness User

Key comments

- Needs guidance and confidence when exercising.

Pain points

- Does not know which exercises are suitable.
- Feels uncomfortable exercising alone.

Feature suggestions

- Beginner-friendly AI coaching.
- Community support groups.
- Trainer recommendations.

Participant P04 – Fitness Trainer

Key comments

- Users need realistic fitness goals.
- Many beginners lose motivation quickly.

Pain points

- Generic workout recommendations.
- Lack of user accountability.

Feature suggestions

- Progress monitoring.
- Trainer-supported communities.
- Personalised exercise suggestions.

Participant P05 – Gym Instructor

Key comments

- Nutrition tracking is one of the biggest challenges.

Pain points

- Manual food logging is time-consuming.
- Users stop tracking after several days.

Feature suggestions

- Camera-based food recognition.
- Automatic calorie estimation.
- A simple nutrition dashboard.

Key Research Findings

Finding	Evidence
Personalisation gap	Users disliked generic workout plans.
Lack of motivation	Users wanted challenges and achievements.
Social isolation	Users requested communities and accountability partners.
Nutrition-tracking difficulty	Manual food logging was considered tedious.
Need for flexibility	Busy users required adaptive workout schedules.

Activity 6: Prepare Consent Forms and Retain Raw Data

Participant Consent Form

FitFlow App Redesign – User Research Study

Study Purpose

The purpose of this study is to understand user experiences, challenges, and expectations regarding the FitFlow fitness-tracking application. The findings will be used to improve usability and design new features such as AI-powered workouts, social communities, and improved nutrition tracking.

Participation Information

- Participation in this study is completely voluntary.
- Participants may answer interview questions, complete surveys, or take part in observation sessions.
- The study will take approximately 15–30 minutes.

Data Collection

The following information may be collected:

- ☐ Survey responses
- ☐ Interview notes
- ☐ Audio recordings (if permission is given)
- ☐ Observation notes

Recording Permission

I understand that:

- ☐ I allow my interview session to be recorded.
- ☐ I do not allow my interview session to be recorded.

Confidentiality

- Participant information will remain confidential.
- Personal identifiers will not appear in research reports.
- Data will be used for academic research purposes only.

Right to Withdraw

Participants may withdraw from the study at any time without giving a reason.

Consent Declaration

I confirm that:

- ☐ I have received information about the study.
- ☐ I understand my participation is voluntary.
- ☐ I agree to participate in the FitFlow user-research study.

Participant Name: _____

Signature: _____

Researcher Signature: _____

Date: _____

Raw Data Storage Plan

Data Type	Storage Method
Interview notes	Stored in encrypted digital documents.
Survey responses	Stored in a password-protected spreadsheet.
Audio recordings	Stored securely and deleted after analysis.
Observation notes	Stored in the research documentation folder.
Consent forms	Stored separately from the research data.

Data Anonymisation Strategy

To protect participant privacy:

- Participants are assigned IDs such as P01, P02, and P03.
- Names and personal information are not included in reports.
- Findings present summarised information only.
- Direct quotes do not contain identifying details.

Data Security Measures

The research team will:

- Store files in password-protected storage.
- Restrict access to authorised researchers only.
- Avoid sharing personal participant information.
- Follow privacy standards such as GDPR and CCPA requirements.
- Delete unnecessary raw data after project completion.